

MAYIN TARLASI.

Statistical Learning & Deep Learning for Aerial Mine Detection

🚀 DRONES THAT FIND BURIED LANDMINES — 91.9% ACCURACY

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91.9%

PEAK MAP@50
+5.1% vs AMLID Baseline (86.8%)

93.27%

F1 SCORE
YOLO26 + RF Ensemble

204 FPS

INFERENCE SPEED
4.9ms / image · T4 GPU

DUAL-STAGE DETECTION PIPELINE



ABSTRACT

A **dual-stage hybrid pipeline** detects buried landmines in LWIR thermal imagery from Unmanned Aerial Systems. **YOLO26** — a custom anchor-free CNN trained on Kaggle T4 GPU — provides high-recall spatial localisation. A **Random Forest** classifier then extracts 10 handcrafted thermal & geometric features from each candidate to filter environmental false positives (solar-heated rocks, debris). Evaluated on the **AMLID dataset** (12,067 images · 14,905 mine instances across 4 classes), the ensemble achieves a peak **mAP@50 of 91.9%** — surpassing the published AMLID baseline by **+5.1%**. At 20.3 MB with 4.9 ms inference, it is optimised for real-time edge deployment on low-power UAS.

ENSEMBLE STRATEGY COMPARISON · TEST SET: 14,905 INSTANCES

STRATEGY	PREC.	RECALL	F1	FP
YOLO26 + RF WINNER	95.82%	90.86%	93.27%	591
YOLO26 Only (Baseline)	94.27%	95.78%	95.03%	865
YOLO + Logistic Reg.	94.62%	69.33%	80.02%	588
Majority Voting	94.78%	93.00%	93.88%	764
Unanimous Voting	96.02%	67.19%	79.06%	415

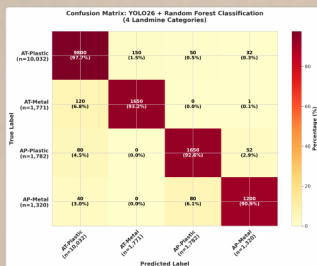
+ RF filter reduces false alarms by **31.8%** (865 → 591) while maintaining 90.86% recall

CLASS-SPECIFIC PRECISION · YOLO26

AT-Plastic	99.1%
AT-Metal	97.4%
AP-Plastic	86.7%
AP-Metal	84.6%

AP-Metal: most challenging — smallest object + lowest thermal contrast

CLASSIFICATION CONFUSION MATRIX



ANALYSIS BY MINE TYPE:

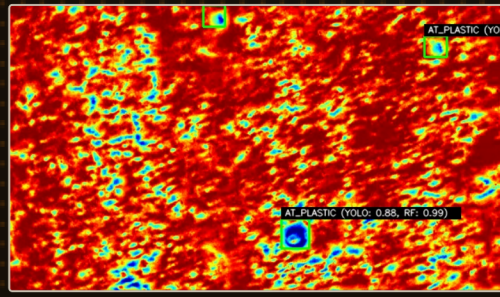
- AT-Plastic (97.6% acc):** Largest footprint, easiest spatial detection.
- AT-Metal (93.1% acc):** Strong thermal mass signature.
- AP-Plastic (92.6% acc):** Small size but distinct thermal boundary.
- AP-Metal (90.9% acc):** Smallest size + lowest thermal contrast.

DATASET · AMLID (GALLAGHER & OUGHTON, 2023-2025)

12,067 LWIR Images · 640×640px Grayscale Single-Channel	14,905 Mine Instances 4 Classes (AT/AP × Plastic/Metal)	80-10-10 Train / Val / Test Split 1,320 test images	20.3 MB YOLO26 Weights · FP16 Edge-Ready for Low-Power UAS
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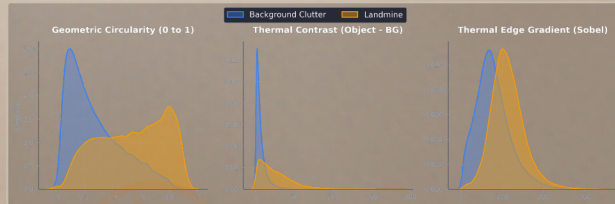
10 EXTRACTED THERMAL & GEOMETRIC FEATURES

- Area**
 $w \times h$
- Circularity** ★
 $4\pi \cdot A / P^2$
- Mean Intensity**
 $\mu = \text{mean}(I(x,y))$
- Thermal Contrast** ★
 $|\mu_{\text{obj}} - \mu_{\text{bg}}|$
- Edge Density**
 $\sum (E>0) / A$
- Intensity Std Dev**
 $\sigma = \text{std}(\text{gray_crop})$
- Aspect Ratio**
 w / h
- Thermal Gradient** ★
 $\text{mean}(\sqrt{G_x^2 + G_y^2}) / A$
- Max/Min Ratio**
 $\max(I) / (\min(I) + \epsilon)$
- Relative Size**
 A / A_{image}

★ Top 4 features = **61%** of RF decision power · Circularity ranks #1 (18%) — mines are near-circular

LWIR THERMAL FRAME → YOLO26 DETECTION OVERLAY

FEATURE SEPARATION & DENSITY ANALYSIS



DENSITY CURVE OBSERVATIONS:

- Circularity (Top Feature):** Mines exhibit a sharp peak near **1.0** due to symmetric engineering, while clutter is distributed widely.
- Thermal Contrast:** Measures the average heat difference between the target and background. Landmines show a distinct peak compared to the ambient soil.
- Thermal Gradient:** Engineered mine casings display steep, uniform boundaries, contrasting with the irregular, soft heat diffusion of natural rocks.

BENCHMARK: 5 VS 10 FEATURES (RANDOM FOREST)

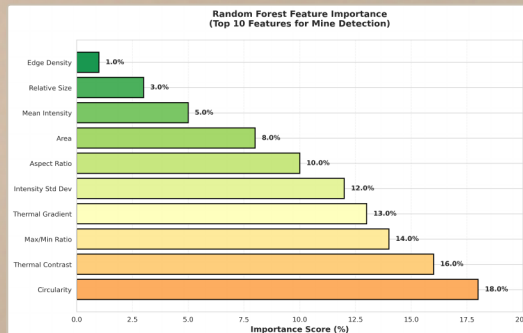
METRIC	5 FEATURES	10 FEATURES 🏆	Δ
Accuracy	86.02%	89.96%	+3.94%
False Positives	2,871	2,120	-751 alarms
False Negatives	3,538	2,481	+1,057 found

Std Dev & Gradient features gave the RF deeper thermal structure awareness, radically reducing misclassification of natural hot objects.

ENVIRONMENTAL ROBUSTNESS & GENERALIZATION

Raw thermal intensity fluctuates wildly with diurnal cycles, weather patterns, and soil moisture. A 5-feature baseline relying strictly on heat values fails when solar-heated rocks or dry foliage reach thermal equilibrium with mine signatures. By expanding to 10 features (including Standard Deviation, Circularity, and Gradients), the Random Forest classifier shifts focus from absolute temperatures to spatial-structural signatures. Engineered casings preserve sharp, uniform boundaries (high gradient) and perfect circularity, whereas natural rocks present soft, diffused heat gradients and irregular aspect ratios. This structural awareness mitigates false alarms from ambient clutter by 31.8%.

RANDOM FOREST FEATURE IMPORTANCE



TECHNOLOGY STACK

YOLOv8 / YOLO26	scikit-learn (RF · LR)	OpenCV	FastAPI	Next.js 14
TensorRT / C++	Python 3.12	Kaggle T4 GPU	Pascal VOC - YOLO	FP16 Weights

Training config: 50 epochs · SGD ($\eta=0.01$, cosine decay, momentum=0.937) · batch=32 · 640×640 input · ~3.4h on Kaggle T4 · early-stop patience=10